

Name:

BUID:

More Practice Problems

1.

$$A = \begin{bmatrix} 0 & 2 & 0 & 4 & 1 \\ 0 & -3 & 1 & -3 & 0 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 3 & 0 \end{bmatrix}$$

- (a) Determine a basis for $\text{Col } A$ (b) Determine a basis for $\text{Nul } A$
(c) Express $\mathbf{0}$ as a nontrivial linear combination of the columns of A
(d) If A is interpreted as the augmented matrix of a linear system, does this system have a unique least squares solution?

2. Let A be a 7×5 matrix such that

$$A^T A = \begin{bmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- (a) Determine the singular values of A (b) Determine $\dim(\text{Nul } A)$ using the fact that $\text{Nul } A = \text{Nul } A^T A$
(c) Determine $\text{rank } A$ (d) Determine $\text{rank } A^T$ and $\dim(\text{Nul } A^T)$
(e) Determine V in the SVD of A (assuming nonincreasing order of singular values in Σ)

3.

$$\mathcal{B} = \left\{ \begin{bmatrix} 3 \\ 7 \end{bmatrix}, \begin{bmatrix} -2 \\ -4 \end{bmatrix} \right\}$$

- (a) Determine the matrix that implements the change of basis matrix for $\mathcal{B}$
(b) Determine an orthonormal basis of $\mathbb{R}^2$ which contains a scalar multiple of $\begin{bmatrix} -2 \\ -4 \end{bmatrix}$

~~4. Let A be a symmetric matrix such that $\text{Nul}(A) = \text{span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \right\}$ and $A \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ -2 \end{bmatrix}$. Determine A .~~

Write your solutions below and on the back on this page.

1.
$$\begin{bmatrix} 0 & 2 & 0 & 4 & 1 \\ 0 & -3 & 1 & -3 & 0 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 3 & 0 \end{bmatrix}$$

 $\vec{a}_1 \quad \vec{a}_2 \quad \vec{a}_3 \quad \vec{a}_4 \quad \vec{a}_5$

$$\text{Col}(A) = \text{span} \{ \vec{a}_2, \vec{a}_3, \vec{a}_5 \}$$

$$\text{Nul}(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 3 \\ -1 \\ 0 \end{bmatrix} \right\}$$

$$\vec{a}_4 = 2\vec{a}_2 + 3\vec{a}_3 \Rightarrow 2\vec{a}_2 + 3\vec{a}_3 - \vec{a}_4 = \vec{0}$$

2. No. $\{ \vec{a}_1, \vec{a}_2, \vec{a}_3, \vec{a}_4 \}$ is L.D.

Continue your work here.

$$2. \textcircled{a} \begin{bmatrix} 2-\lambda & -1 & 0 & 0 & 0 \\ -1 & 2-\lambda & 0 & 0 & 0 \\ 0 & 0 & 4-\lambda & 0 & 0 \\ 0 & 0 & 0 & -\lambda & 0 \\ 0 & 0 & 0 & 0 & -\lambda \end{bmatrix} \sim \begin{bmatrix} 2-\lambda & -1 & 0 & 0 & 0 \\ -(2-\lambda) & (2-\lambda)^2 & 0 & 0 & 0 \\ 0 & 0 & 4-\lambda & 0 & 0 \\ 0 & 0 & 0 & -\lambda & 0 \\ 0 & 0 & 0 & 0 & -\lambda \end{bmatrix} \sim \begin{bmatrix} 2-\lambda & -1 & 0 & 0 & 0 \\ 0 & (2-\lambda)^2-1 & 0 & 0 & 0 \\ 0 & 0 & 4-\lambda & 0 & 0 \\ 0 & 0 & 0 & -\lambda & 0 \\ 0 & 0 & 0 & 0 & -\lambda \end{bmatrix}$$

$$\det(A^T A - \lambda I) = \frac{1}{\cancel{2-\lambda}} (\cancel{2-\lambda})(\cancel{2-\lambda})^2 - 1)(4-\lambda) \lambda^2$$

$$= -(\lambda^2 - 4\lambda + 4 - 1)(\lambda - 4) \lambda^2$$

$$= -(\lambda - 3)(\lambda - 1)(\lambda - 4) \lambda^2$$

$$\sigma_1 = 2 \quad \sigma_2 = \sqrt{3} \quad \sigma_3 = 1, \quad \sigma_4 = 0$$

$$\textcircled{b} \dim(\text{Nul } A) = \dim(\text{Nul } (A^T A)) = 2$$

$$\textcircled{c} \text{rank } A = 5 - \dim(\text{Nul } A) = 3$$

$$\textcircled{d} \text{rank}(A^T) = \text{rank}(A) = 3 \quad \dim(\text{Nul } A^T) = 4$$

$$\textcircled{e} V = \begin{bmatrix} 0 & r_{1/2} & r_{1/2} & 0 & 0 \\ 0 & -r_{1/2} & r_{1/2} & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$3. \textcircled{a} \begin{bmatrix} 3 & -2 \\ 7 & -4 \end{bmatrix}^{-1} = \frac{1}{-12+14} \begin{bmatrix} -4 & 2 \\ -7 & 3 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ -7/2 & 3/2 \end{bmatrix}$$

$$\textcircled{b} \vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \quad B = \left\{ \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} \right\}$$